Decision Letter for Sumbission M187198

Editor (Remarks to the Author):

The manuscript proposes a stochastic primal-dual framework for deep hashing with W-type quantization regularization, together with convergence/complexity analyses and an extensive experimental evaluation. Both reviewers acknowledge positive aspects of the work, in particular the experimental section, which is comprehensive and demonstrates the practical potential of the proposed approach.

However, both reviews (see attached files) also raise **substantial concerns regarding the theoretical contribution**, which is central to the manuscript.

First, the connection between the proposed primal-dual reformulation and the original nonconvex W-regularized hashing problem is not established with sufficient rigor. In particular, the manuscript does not clearly characterize the duality or stationarity relationship between the two formulations, making it difficult to assess what guarantees obtained for the reformulated problem imply for the original optimization problem. The scope of the claimed "W-type" regularization framework and some elements of the problem reformulation also require significantly clearer definitions.

Second, the detailed examination of the convergence and complexity analysis reveals several mathematical issues that go beyond presentation. These include missing or insufficient assumptions, incomplete boundedness and descent arguments, an insufficiently formalized stochastic setting, and errors in the manipulation of conditional expectations and in some estimates. Importantly, some of these issues propagate to the main complexity results, so that the theoretical guarantees claimed in the current manuscript cannot be considered established as written.

Decision Letter

Dear Dr. Cui,

Thank you for your manuscript, "DualHash: A Stochastic Primal-Dual Algorithm with Theoretical Guarantee for Deep Hashing," submitted for possible publication to SIAM Journal on Imaging Sciences.

I regret to inform you that though your manuscript is interesting, we will not accept it for publication.

SIAM Journal on Imaging Sciences aims to publish the most significant work going on in the mathematical and formal aspects of imaging sciences. The results of your paper

are not of sufficient mathematical interest or significance for publication in a flagship journal such as SIAM Journal on Imaging Sciences.

Best regards